

Problem

Simplify the following expressions:

(a)
$$\frac{(5 - j6) - (2 + j8)}{(-3 + j4)(5 - j) + (4 - j6)}$$

(b)
$$\frac{(240 \angle 75^\circ + 160 \angle -30^\circ)(60 - j80)}{(67 + j84)(20 \angle 32^\circ)}$$

(c)
$$\left(\frac{10 + j20}{3 + j4} \right)^2 \sqrt{(10 + j5)(16 - j20)}$$

Step-by-step solution

Step 1 of 4

(a)

The following is the given expression:

$$\frac{(5 - j6) - (2 + j8)}{(-3 + j4)(5 - j) + (4 - j6)}$$

Simplify this expression.

$$\begin{aligned} &\Rightarrow \frac{(5 - j6) - (2 + j8)}{(-3 + j4)(5 - j) + (4 - j6)} \\ &\Rightarrow \frac{5 - j6 - 2 - j8}{(-15 + j3 + j20 + 4) + (4 - j6)} \\ &\Rightarrow \frac{(5 - 2) - (j6 + j8)}{(-15 + 4 + 4) + (j3 + j20 - j6)} \end{aligned}$$

Rationalize the expression.

$$\begin{aligned} &\Rightarrow \left(\frac{3 - j14}{-7 + j17} \right) \left(\frac{-7 - j17}{-7 - j17} \right) \\ &\Rightarrow \frac{-21 + j98 - j51 - 238}{49 + j119 - j119 + 289} \\ &\Rightarrow \frac{-259 + j47}{338} \\ &\Rightarrow -0.766 + j0.139 \end{aligned}$$

Therefore, the simplified expression is $0.765 - j0.139$.

[Comments \(1\)](#)



Anonymous

signs seem to be reversed in final solution

Step 2 of 4

(b)

The following is the given expression:

$$\frac{(240 \angle 75^\circ + 160 \angle -30^\circ)(60 - j80)}{(67 + j84)(20 \angle 32^\circ)}$$

Simplify this expression.

$$\begin{aligned} & \frac{(240 \angle 75^\circ + 160 \angle -30^\circ)(60 - j80)}{(67 + j84)(20 \angle 32^\circ)} \\ \Rightarrow & \frac{(62.12 + j231.82 + 138.56 - j80)(60 - j80)}{(67 + j84)(16.96 + j10.59)} \\ \Rightarrow & \frac{(200.68 + j151.82)(60 - j80)}{(67 + j84)(16.96 + j10.59)} \\ \Rightarrow & \frac{24186.4 - j6945.2}{246.76 + j2134.17} \end{aligned}$$

Rationalize the expression.

$$\begin{aligned} \Rightarrow & \frac{(24186.4 - j6945.2)(246.76 - j2134.17)}{(246.76 + j2134.17)(246.76 - j2134.17)} \\ \Rightarrow & \frac{-8854001.42 - j53331686.84}{4615572.08} \\ \Rightarrow & -1.92 - j11.55 \end{aligned}$$

Therefore, the simplified expression is $\boxed{-1.92 - j11.55}$.

Step 3 of 4

(c)

The following is the given expression:

$$\left(\frac{10 + j20}{3 + j4} \right)^2 \sqrt{(10 + j5)(16 - j20)}$$

Simplify this expression.

$$\begin{aligned} & \left(\frac{10 + j20}{3 + j4} \right)^2 \sqrt{(10 + j5)(16 - j20)} \\ \Rightarrow & \left(\frac{-300 + j400}{-7 + j24} \right) \sqrt{260 - j120} \\ \Rightarrow & \left(\frac{-300 + j400}{-7 + j24} \right) \left(\frac{-7 - j24}{-7 - j24} \right) \sqrt{260 - j120} \\ \Rightarrow & \left(\frac{11700 + j4400}{625} \right) \sqrt{260 - j120} \end{aligned}$$

Step 4 of 4

On further simplification:

$$\Rightarrow (18.72 + j7.04) \sqrt{\sqrt{(260)^2 + (120)^2} \angle \tan^{-1} \left(\frac{-120}{260} \right)}$$

$$\Rightarrow (18.72 + j7.04) \sqrt{\sqrt{82000} \angle \tan^{-1} (0.8)}$$

$$\Rightarrow (18.72 + j7.04) \sqrt{286.35 \angle -24.77^\circ}$$

$$\Rightarrow (18.72 + j7.04) (16.92 \angle -12.38^\circ)$$

$$\Rightarrow (18.72 + j7.04) (16.53 - j3.63)$$

$$\Rightarrow 335 + j48.42$$

Therefore, the simplified expression is $\boxed{335 + j48.42}$.